



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ARUNA FUEL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

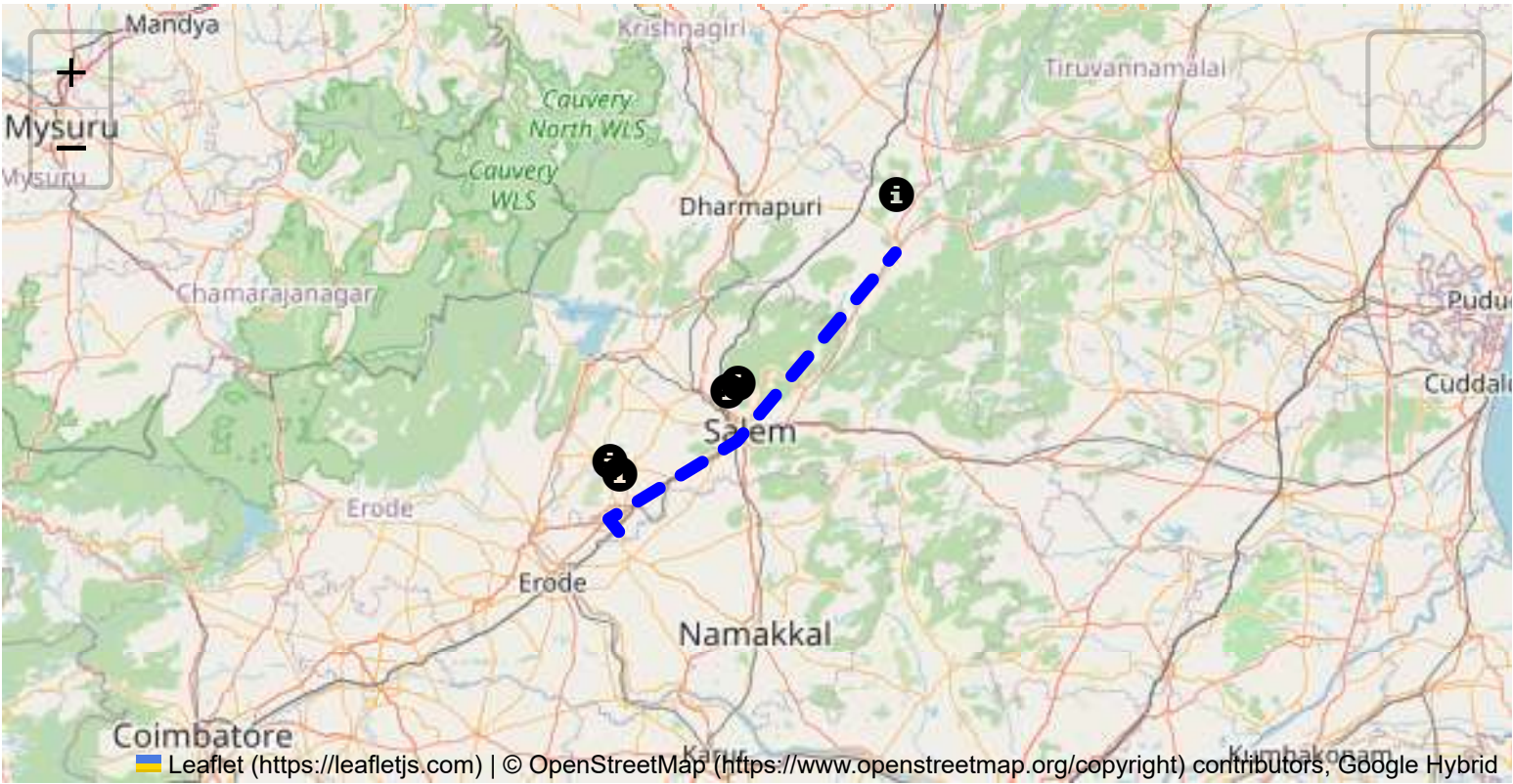
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

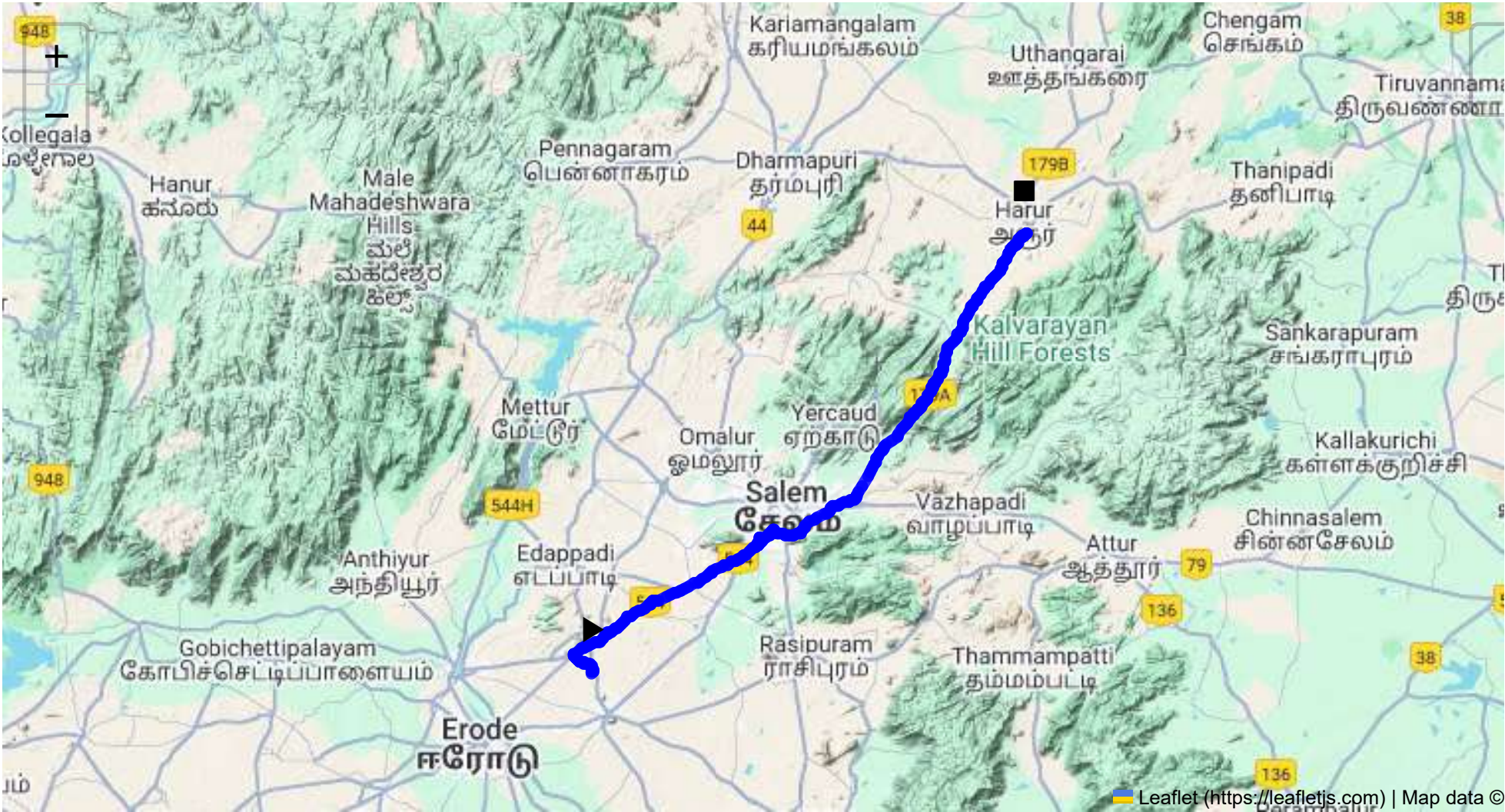
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.66 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.032627, 78.473362)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route starts from CVQF+23W, Sangagiri, Tamil Nadu, and ends at 2FMF+38 Dodampatty, Tamil Nadu. It passes through key waypoints including FR7X+7V9, Tamil Nadu 637303, 110 Uthamasolapuram, and J4MH+239, P M P Nagar. The route comprises a mix of highways and local roads, with varied traffic densities and infrastructure conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences hot and humid conditions, with monsoon rains from June to September. During the monsoon, roads may become slippery, and low-lying areas might face temporary flooding, leading to delays and potentially hazardous driving conditions.

3. Analysis of Traffic Patterns

Traffic is generally busiest during the morning (8:00-10:00 AM) and evening (5:00-7:00 PM) peak hours, especially around urban areas such as Salem. Congestion is common in these areas, particularly around major junctions and when passing through city centers.

4. Assessment of Road Quality and Infrastructure

The initial segment near Sangagiri features well-maintained highways. However, as you move towards Salem and beyond, road quality may deteriorate, with some sections having potholes or undergoing construction. The urban stretches, especially around Salem, may have narrower lanes and more frequent intersections.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, consider the following alternatives:

- Taking NH544 can offer a more direct route while bypassing some congested urban areas.
- Small rural roads can detour around blocked areas but may be slower and not well-suited for heavy trucks.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires compliance with specific safety and documentation standards. Stay updated on local bylaws regarding permitted hours for such movements and ensure that all materials are properly classified, packaged, and labeled.

7. Overview of Historical Incidents

Salem has seen a few incidents involving heavy vehicles, often due to brake failures or driver errors on steep gradients. Past incidents highlight the need for frequent vehicle checks and driver awareness programs.

8. Environmental Considerations and Sensitive Areas

The route skirts around several agricultural patches and water bodies. Minimize ecological disturbance by strictly adhering to designated routes and avoiding unauthorized stops in sensitive areas.

Mismanagement can lead to contamination risks and attract penalties.

9. Analysis of Communication Coverage

Urban regions generally have good mobile network coverage. However, rural stretches might experience weak or no signal, potentially affecting emergency communication. Keep alternative communication methods handy.

10. Estimated Emergency Response Times

In urban stretches like Salem, response times are quicker, typically within 15-30 minutes. More remote areas could see a delay of 45-60 minutes. Staying informed about local emergency services can help improve response coordination.

12. Overall Summary of Risk Assessment

This route poses moderate risks primarily due to traffic congestion, road conditions, and inclement weather during monsoons. Special attention should be placed on night driving and rural areas with limited communication. By adhering to local laws and being prepared for sudden changes in weather and road conditions, hazardous material transport can proceed safely. It is recommended to regularly update drivers on route conditions and ensure vehicles are well-maintained and stocked with safety equipment.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	U-Turn	High	12.0372299, 78.47784910000001	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45853, 77.85139	30 KM/Hr
7	Turn	Medium	11.45906, 77.85132	30 KM/Hr
8	Turn	High	11.46332, 77.84935	15 KM/Hr
9	Turn	Medium	11.47358, 77.86766	30 KM/Hr
10	Turn	High	11.49150, 77.89602	15 KM/Hr
11	Turn	High	11.49224, 77.89611	15 KM/Hr
12	Turn	Medium	11.53584, 77.95814	30 KM/Hr
13	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
14	Turn	High	11.62130, 78.10776	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
16	Turn	Medium	11.61224, 78.10469	30 KM/Hr
17	Blind Spot	Blind Spot	11.63306, 78.12722	10 KM/Hr
18	Turn	Medium	11.62826, 78.13337	30 KM/Hr
19	Turn	Medium	11.62836, 78.16483	30 KM/Hr
20	Turn	High	11.66380, 78.20648	15 KM/Hr
21	Turn	High	11.67120, 78.23136	15 KM/Hr
22	Turn	High	11.67190, 78.23144	15 KM/Hr
23	Turn	Medium	11.67429, 78.23896	30 KM/Hr
24	Turn	Medium	11.81219, 78.34077	30 KM/Hr
25	Turn	Medium	11.83472, 78.35285	30 KM/Hr
26	Turn	Medium	11.99572, 78.44559	30 KM/Hr
27	Turn	Medium	12.01548, 78.45650	30 KM/Hr
28	Blind Spot	Blind Spot	12.03717, 78.47794	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
1	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
2	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
3	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
4	hospital	SP Hospital	11.4708567, 77.8654872	30 km/h	Medium
5	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
7	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
9	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
10	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
12	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
13	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
15	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
16	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
17	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
18	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
19	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
21	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
26	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
27	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
29	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
30	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
31	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
32	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
33	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
34	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
35	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
37	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
38	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
39	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
42	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
43	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
44	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
11	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
20	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
22	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
23	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
24	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
25	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
28	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
40	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
41	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
45	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
46	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
47	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
48	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
49	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
50	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 12.0372299, 78.47784910000001



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45353, 77.85697



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45853, 77.85139



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45906, 77.85132



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46332, 77.84935



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47358, 77.86766



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49150, 77.89602



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53584, 77.95814



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.62145, 78.10740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.61235, 78.10438



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61224, 78.10469



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63306, 78.12722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62836, 78.16483



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66380, 78.20648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67190, 78.23144



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81219, 78.34077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83472, 78.35285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99572, 78.44559



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01548, 78.45650



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.03717, 78.47794

